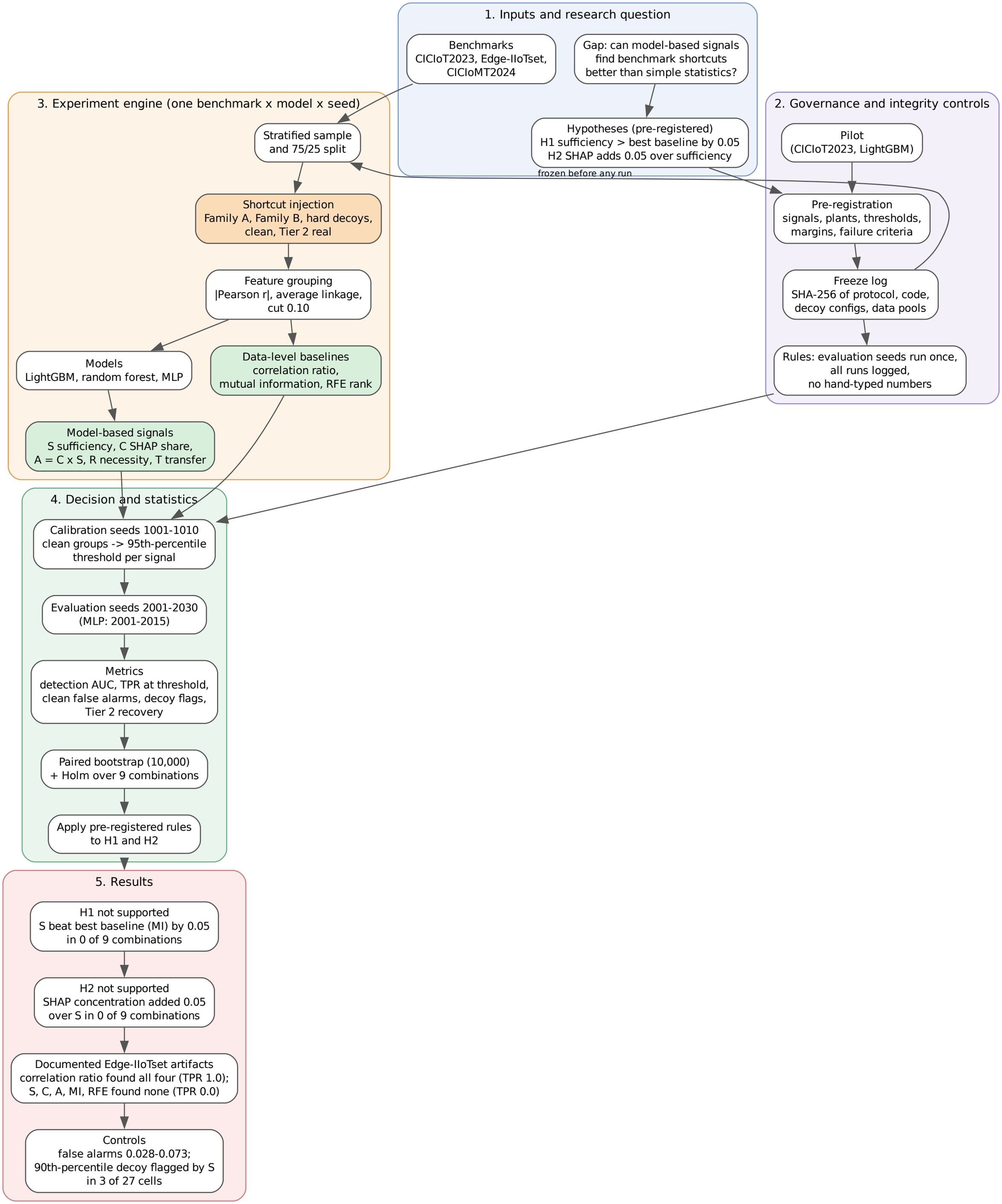
1. Inputs and research question
Benchmarks
CICIoT2023, Edge-IIoTset,
CICIoMT2024
Gap: can model-based signals
find benchmark shortcuts
better than simple statistics?
Hypotheses (pre-registered)
H1 sufficiency > best baseline by 0.05
H2 SHAP adds 0.05 over sufficiency
frozen before any run
2. Governance and integrity controls
Pilot
(CICIoT2023, LightGBM)
Pre-registration
signals, plants, thresholds,
margins, failure criteria
Freeze log
SHA-256 of protocol, code,
decoy configs, data pools
Rules: evaluation seeds run once,
all runs logged,
no hand-typed numbers
3. Experiment engine (one benchmark x model x seed)
Stratified sample
and 75/25 split
Shortcut injection
Family A, Family B, hard decoys,
clean, Tier 2 real
Feature grouping
|Pearson r|, average linkage,
cut 0.10
Models
LightGBM, random forest, MLP
Data-level baselines
correlation ratio,
mutual information, RFE rank
Model-based signals
S sufficiency, C SHAP share,
A = C x S, R necessity, T transfer
4. Decision and statistics
Calibration seeds 1001-1010
clean groups -> 95th-percentile
threshold per signal
Evaluation seeds 2001-2030
(MLP: 2001-2015)
Metrics
detection AUC, TPR at threshold,
clean false alarms, decoy flags,
Tier 2 recovery
Paired bootstrap (10,000)
+ Holm over 9 combinations
Apply pre-registered rules
to H1 and H2
5. Results
H1 not supported
S beat best baseline (MI) by 0.05
in 0 of 9 combinations
H2 not supported
SHAP concentration added 0.05
over S in 0 of 9 combinations
Documented Edge-IIoTset artifacts
correlation ratio found all four (TPR 1.0);
S, C, A, MI, RFE found none (TPR 0.0)
Controls
false alarms 0.028-0.073;
90th-percentile decoy flagged by S
in 3 of 27 cells